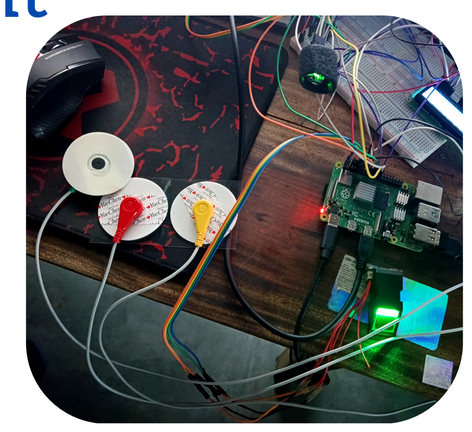
Heart Attack Detection in IoT Environment

When Your Heart Speaks, Our Device Responds

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Introduction

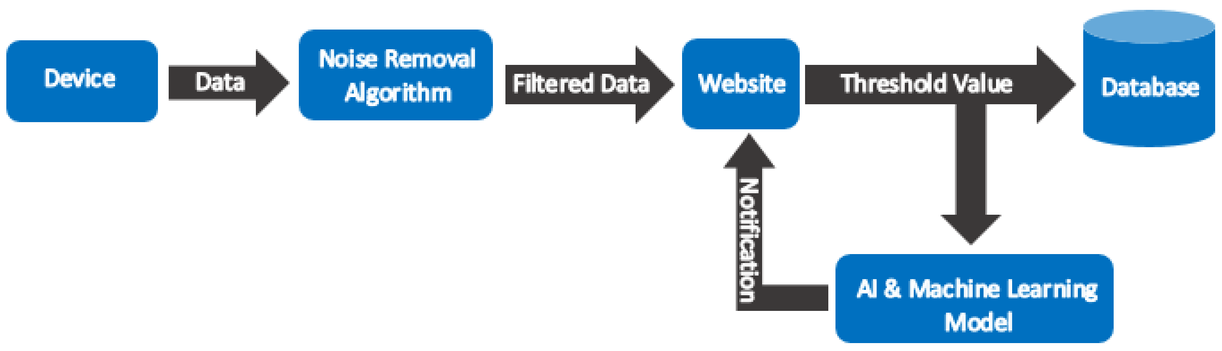
Heart attacks can happen without warning. Quick detection is critical to save lives. Our project uses IoT technology to monitor heart rate in real-time and send alerts when abnormal patterns are detected. It's a smart solution for early warning and better care.

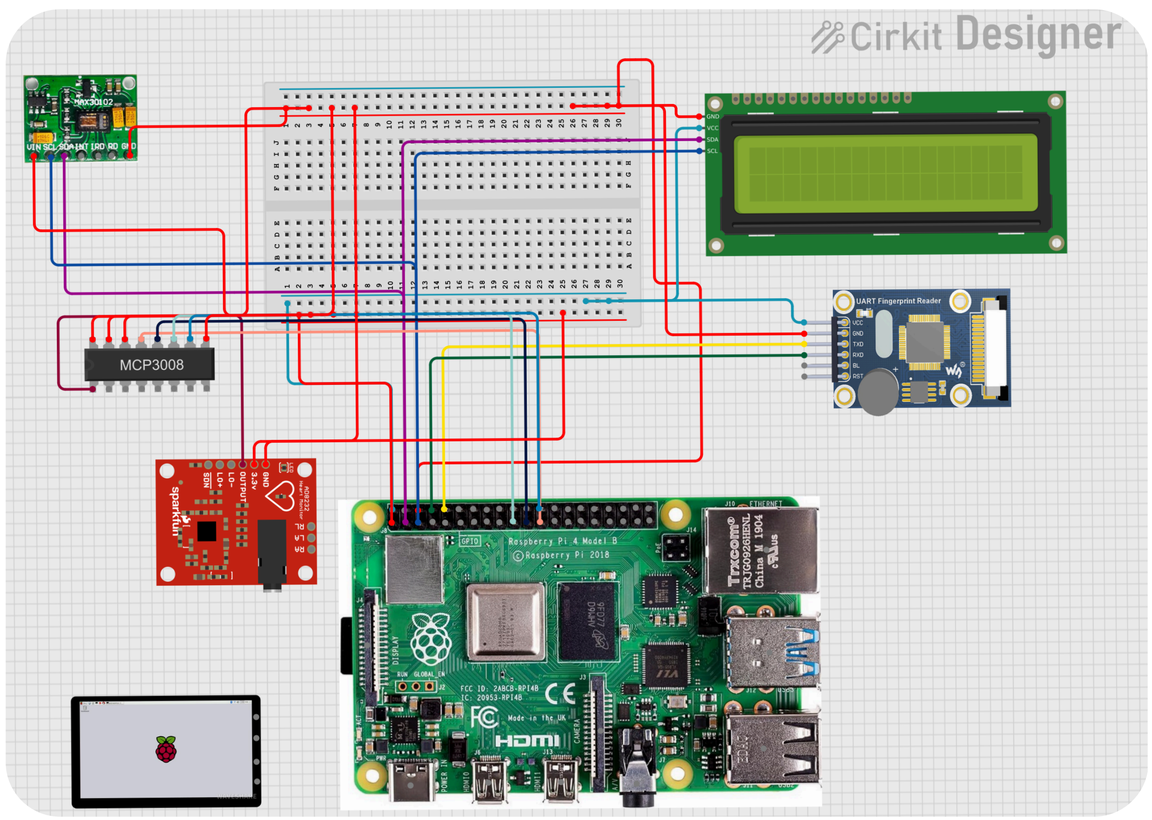


Objectives

- To monitor heart rate in IoT technology that detect abnormal heart patterns and send instant alerts to consultant doctor/care viewer.
- To enable remote health monitoring through a website that reduces diagnosis delay after a heart attack.
- To ensure better security with our compact, portable and affordable system.

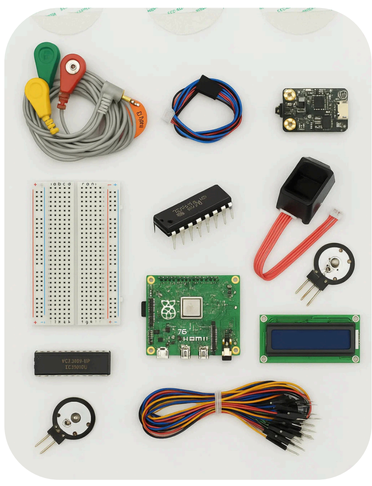
Methodology





System Components

- Raspberry Pi 4
- ECG Sensor
- Fingerprint Sensor
- Pulse Sensor
- MCP 3008 ADC
- 16 x 2 Display
- Breadboard
- Jumper Wire



Challenges

📊Data Accuracy: Ensuring precise and consistent heart rate data in real-time .

🧩Component Integration: Seamlessly connecting with multiple hardware modules .

🌐Website Synchronization: Ensuring smooth and timely data updates in the database.

🧠Real-time Processing: Managing fast data handling & decision-making on limited hardware .

🔐Data Security: Protecting patient health data during wireless transmission .

Conclusion

Our system is more than a device — it's a life-saving companion. With real-time heart monitoring and smart IoT alerts, it brings technology closer to care, ensuring that no heartbeat goes unheard when it matters most.

Future Scope

🧠AI-based Analysis to predict heart conditions before they become critical.

🛡️Enhanced Security to protect sensitive health data over wireless networks .

References

[1] WHO,"cardiovascular diseases"(https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases)

[2] Cirkit Designer (https://www.cirkitstudio.com/)

Web UI & Expected output

